



DrillMain Intelligence

Smart drilling insights with next-step guidance and integrated maintenance

Table of Contents

Explore the journey of DrillMain Intelligence from concept to impact.



Project Context



Project Objectives



Data Processing Flow



Platform Outcomes



Problem Statement



System Architecture



Challenges & Solutions



Lessons & Future Directions

Project Context

An intelligent drilling operations platform that transforms daily drilling report data into actionable insights at segment, well, and fleet levels.

The system integrates report ingestion, KPI dashboards, event severity analysis by depth interval, maintenance risk indicators, fleet-wide reporting and well location mapping.





Problem Statement

Lengthy Reports

Drilling reports are extensive and time-consuming to interpret manually

Delayed Insights

Teams need faster identification of non-productive time and operational issues

Risk Detection

High-risk depth intervals and event severity require clear visualization

Comparative Analysis

Performance comparison across wells and fleet-wide trends are essential

1 General

1.1 Customer Information

Company	OML-11 AMT (NEPL / WAGL)
Representative	MRO ENERGY LIMITED (IPM)
Address	

1.2 Well/Wellbore Information

Well	OKOLOMA-02	Wellbore No.	1
Well Name	OKOLOMA-02	Wellbore Name	OKOLOMA-02
Report no.	25	Report date	14/11/2025
Project	OML - 11	Site	LOCATION 1
Rig Name/No.	L BUBA (ASH)/	Event	ORIG DRILLING
Start date	20/09/2025	End date	
Spud date	23/09/2025	UWI	OKOLOMA-02
Afe No.			
Active datum	DATUM @80.8ft (above Mean Sea Level)		
API no.		Block	OML-11
Supervisor	SUNNY IDOKO / ABDULMALIK ASUBAA.	Engineer	MATTHEW CHUKWUKELU

1.3 Depth Days

MD	6,547.0 (ft)	TVD	
Progress		Avg. ROP	0.0 (ft/hr)
Rotating hours		Cum Rot Hrs	108.67 (hr)
DFS	24.39 (days)	Est days	41.13 (days)
MD auth			

OML-11 AMT (NEPL / WAGL)

2.1 Operation Summary (Continued)

From	To	Dur. (hr)	Phase	Code	Sub	Class	MD from (ft)	MD to (ft)	Operation
									CONT TO WAIT 0N BAKER HUGHES TO PROVIDE FRESH GOOD QUALITY OBM (MUD). MEANWHILE, MONITOR WELL ON TRIP TANK AND BRK CIRC INTERMITTENTLY TO CHECK MWD TOOL. AND CONT RACKING, MEASURING AND DRIFTING 9-5/8" CASING - IN PROGRESS. NPT ON BAKER MUD ENGINEERING.

3 Hole Sections

3.1 Hole Sections

Wellbore Name / Wellbore No.	Section Name	Section type	Effective hole diameter (in)	ID (in)	MD (ft)		Hole section start date/time	Hole section end date/time
					Top	Base		
OKOLOMA-02 / 1	SURFACE	Open Hole	17.500	17.500	0.0	6,030.0	23/09/2025 14:44	04/10/2025 12:45
	12-1/4" DIR HOLE	Open Hole	12.250		12,496.2	6,547.0	11/11/2025 14:00	

4 Fluids

4.1 Mud

Daily mud		0.00 (\$)			Cum. mud cost				0.00 (\$)			
Mud desc.	Density (ppg)	Viscosity (s/qt)	Date/Time	MD check (ft)	PV (cp)	YP (lb/100ft²)	Cl- (ppm)	Ca+ (ppm)	pH	Pm	Pf (cc)	Mf (cc)
SYNTHETIC MUD	9.10	58.00	11/13/2025 0:00		15.00	18.000						

5 Drillstrings

5.1 BHA no.# 3

Date/Time in	11/11/2025 14:00	Date/Time out	
Bit no.	5365561	Weight (Above/Below) Jars	
BHA Length	1,008.05 (ft)	Min. ID	12.250 (in)

1.4 Status

Current status	WAIT ON BAKER MUD ENGINEERING TEAM TO CLEAN UP THE MUD.
24 hr summary	CONT TO WAIT ON BAKER HUGHES TO PROVIDE FRESH GOOD QUALITY OBM (MUD). MEANWHILE, MONITOR WELL ON TRIP TANK AND BRK CIRC INTERMITTENTLY TO CHECK MWD TOOL. AND CONT RACKING, MEASURING AND DRIFTING 9-5/8" CASING - IN PROGRESS.
24 hr forecast	WAIT ON BAKER MUD ENGINEERING TEAM, RECIEVED PRE-MIX, WEIGH UP MUD TO SPECK, RBIH TO BTM & DISPLACE HOLE TO NEW SOBM. DRILLED AHEAD 12-1/4" HOLE APP.

2 Operation Summary

2.1 Operation Summary

From	To	Dur. (hr)	Phase	Code	Sub	Class	MD from (ft)	MD to (ft)	Operation
0:00	0:00	24.00	DRLPRO	21		U	6,547.0	6,547.0	CONT TO WAIT 0N BAKER HUGHES TO PROVIDE FRESH GOOD QUALITY OBM (MUD). MEANWHILE, MONITOR WELL ON TRIP TANK AND BRK CIRC INTERMITTENTLY TO CHECK MWD TOOL. AND CONT RACKING, MEASURING AND DRIFTING 9-5/8" CASING - IN PROGRESS. NPT ON BAKER MUD ENGINEERING. XXXXXXXXXXXXXXXXXXXXX 15/11/2025 XXXXXXXXXXXXXXXXXXXXX 00:00 - 06:00 =====
November 15, 2025 at 6:32 am							1	OpenWells	



OML-11 AMT (NEPL / WAGL)

2.1 Operation Summary (Continued)

From	To	Dur.	Phase	Code	Sub	Class	MD from	MD to	Operation
------	----	------	-------	------	-----	-------	---------	-------	-----------

5.1.2 Assembly Components (Continued)

Component type	No. of joints	Length (ft)	OD (in)	ID (in)	Connection		Weight (ppf)	Grade	Pin Box	Serial no.	Left or right spiral	Fish Neck	
					OD (in)	Name						Length (ft)	OD (in)
Steerable Stabilizer	1	12.12	9.500										
Non-Mag Crossover Sub	1	5.98	9.500										
Logging While Drilling	1	3.60	8.250										
Logging While Drilling	1	8.80	8.250										
Steerable Stabilizer	1	8.74	8.250										
Logging While Drilling	1	6.05	8.250										
Non-Mag Crossover Sub	1	32.56	8.250										
Hang-Off Sub	1	2.10	8.250										
Integral Blade Stabilizer	1	6.32	8.000										
Non-Mag Crossover Sub	1	3.32	8.000										
Float Sub	1	2.37	8.000										
Non-Mag Heavy Weight	1	30.10	8.250										
Circulating Sub	1	9.43	8.000										
Cross Over	1	2.54	8.000										
Heavy Weight Drill Pipe	1	578.03	5.500										
Cross Over	1	2.54	8.000										
Hydraulic Jar	1	31.85	8.000										
Cross Over	1	2.54	8.000										
Heavy Weight Drill Pipe	1	214.49	5.500										

5.1.3 Bit Details

Bit no.	Size (in)	Manufacturer	Model no.	Class	IADC code	Serial no.	Jets (32nd")	MD in (ft)	MD out (ft)	I-O-D-L-B-G-O-R
5365561	12.250	BAKER	TD506K	N		5365561		6,035.0		0-0-NO-A-X- I-NO-

Project Objectives

01

User-Focused Platform

Build intuitive drilling data analysis interface

03

Visual Analytics

Visualize KPIs and depth-segment events clearly

05

Fleet Comparisons

Support fleet-wide comparisons and reporting

02

Data Ingestion

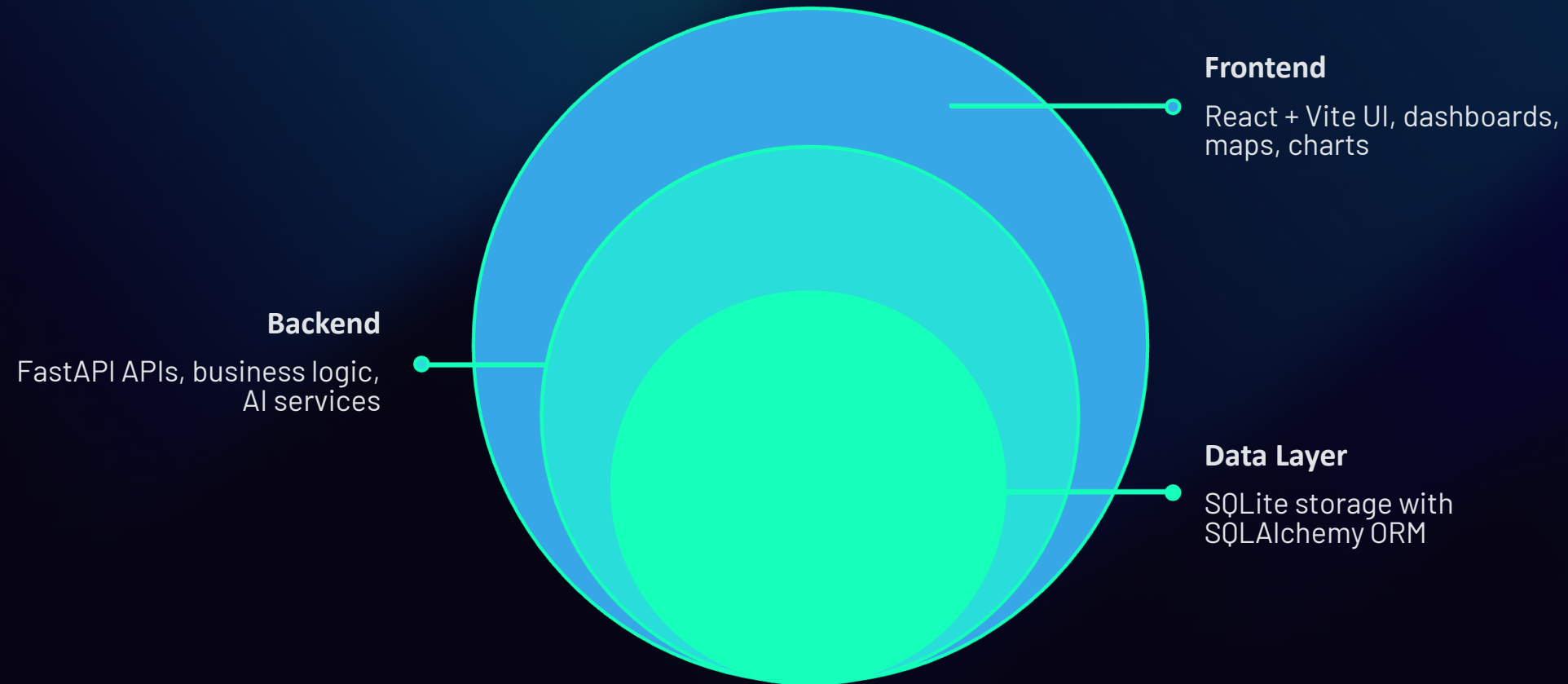
Ingest daily reports and structure operations data

04

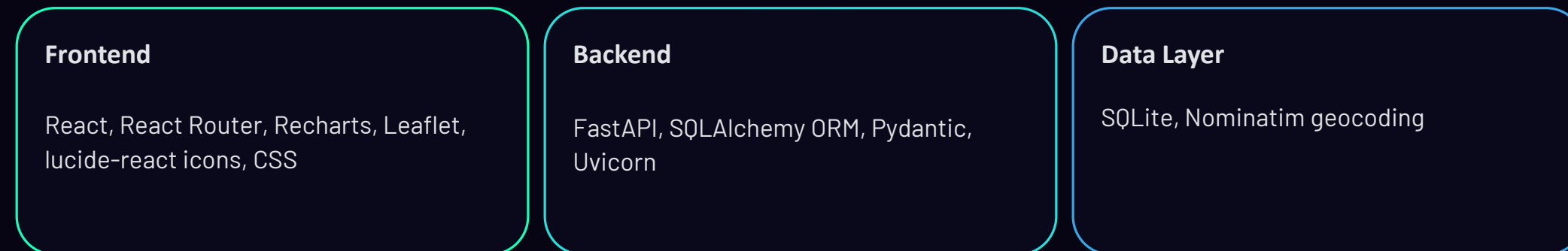
Explainable Analysis

Provide explainable segment-level risk analysis

System Architecture



The architecture is organized in three layers that flow from data storage, through application logic, to the user-facing interface.

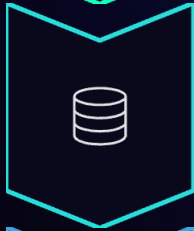


Data Processing Flow



Upload

User uploads drilling report file



Ingestion

Backend extracts operations and related fields



Storage

Records stored in database with structured schema



Aggregation

Dashboard endpoints generate KPI metrics



Analysis

Segment analysis produces risk explanations



Fleet Summary

Multi-well aggregation and reporting

Challenges & Solutions



Slow Fleet Loading

Challenge: Heavy per-well calls caused slow fleet/summary loading

Solution: Implemented timeout strategies, non-blocking requests, and lightweight dashboard mode



Generic AI Explanations

Challenge: Repetitive and generic AI-generated explanations

Solution: Developed description-grounded prompts and rule logic tied to actual segment text



Incomplete PDF parsing

Challenge: Missing details from parsed report.

Solution: Integrated editing method to enable users input information to satisfaction



Platform Outcomes



Faster Insights

Rapid interpretation of drilling reports with automated analysis and clear visualizations



Clear Risk Interpretation

Depth-based risk visualization with severity levels and flagged intervals



Cross-Well Visibility

Fleet-wide performance comparison and operational oversight capabilities

[Back to Dashboard](#)

[Download NPT Report](#)

NPT Analytics — NNPC

Non-Productive Time Analysis



Total NPT

33.0 hrs



Productive Time

909.9 hrs



NPT Percentage

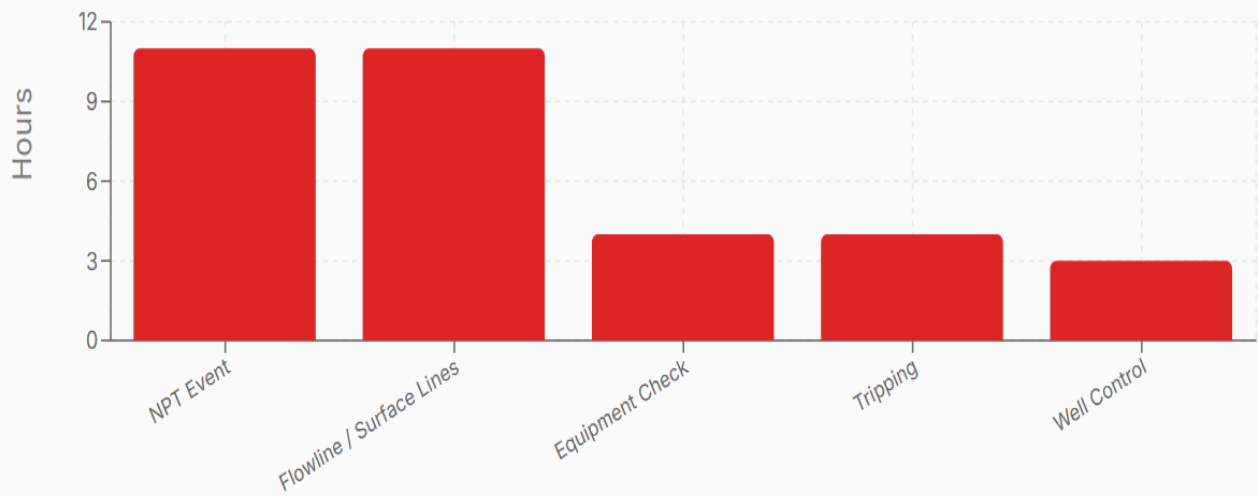
3.50%



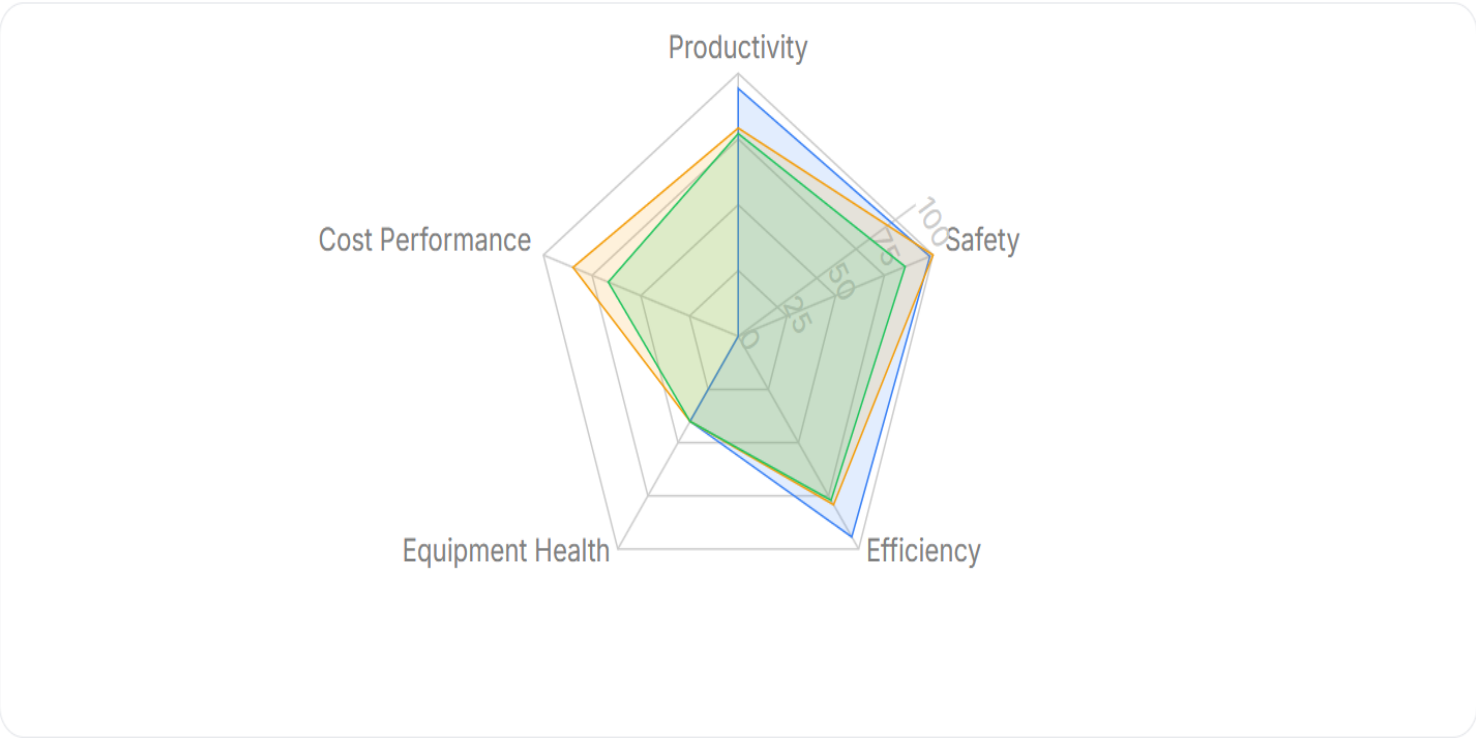
NPT Events

8

NPT by Event Type



Well Performance Comparison (Top 3 Wells)



Detailed Event Breakdown (All Wells)

ID	Well ID	Date	Depth Range	Type	Event	Duration (hrs)	Severity
1	WELL-01	Apr 23	6030-6030m	Warning	Casing / Liner	4.0	Medium
2	WELL-01	Apr 14	11560-11560m	Critical	Casing / Liner	6.0	High
3	Well-02	Apr 23	10326-10326m	Warning	NPT Event	4.0	Medium

Lessons & Future Directions

Key Lessons

- Performance and usability must be designed alongside analytics features
- Explainability is critical for operational trust and adoption
- Small UI decisions significantly impact user adoption rates

Future Enhancements

- Role-based access control and permissions
- Richer report parsing coverage and formats

Project Value: DrillMain Intelligence provides immediate operational value for drilling oversight and risk management, ready for industry application and further research collaboration.

Thank You

We appreciate your time and engagement.

Contact Us

Email: info@drillmain.com

Website: www.drillmain.com

Next Steps

- Schedule a detailed demo